

Plan for this lesson

- Overview of machine learning definitions
- Demo of us using a **chatbot** (ChatGPT) to code
- Demo of us using a **coding agent** (Claude Code) to code
- You try it!

A **machine learning model** is a program that learns patterns from data and uses them to make predictions.



→ dog



→ cat



→ dog



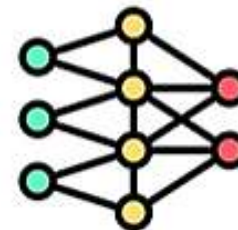
→ cat



→ dog

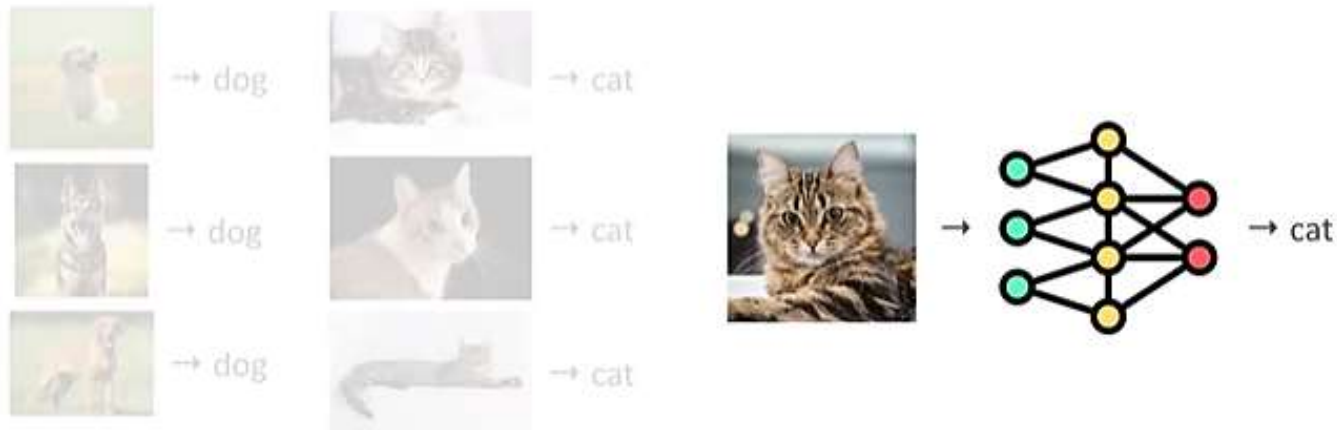


→ cat



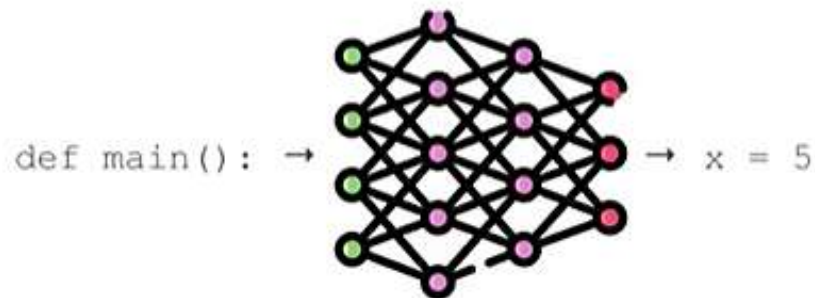


A **machine learning model** is a program that learns patterns from data and uses them to make predictions.





A **Large Language Model (LLM)** is a type of machine learning model trained on huge amounts of text to be able to generate text.



Examples of LLMs:

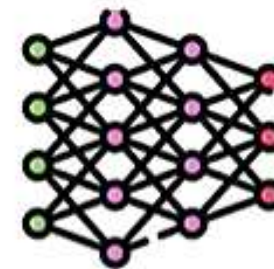
- GPT-5.2
- GPT-5.5
- Claude 3 Opus
- Claude Opus 4.7
- Gemini 3 Pro

You built programs that use GPT



```
main.py
1 from ai import call_gpt
2
3 def main():
4     name = input("Enter your name: ")
5     topic = input("Enter a topic: ")
6     print("Creating your haiku...")
7     print()
8
9     prompt = f"Write a haiku about {name} and {topic}"
10    haiku = call_gpt(prompt)
11    print(haiku)
12
13 if __name__ == "__main__":
14     main()
```

prompt



haiku

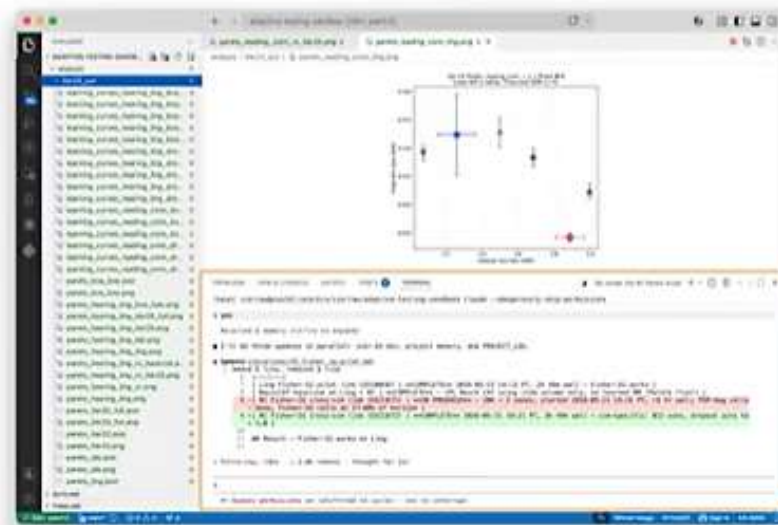
A few different tools



Chatbots



Coding Agents



Autocomplete



ice_cream_sales_dataset

125%

View

Zoom

Add Category

Pivot Table

Insert

Table

Chart

Text

Sh

+

Sheet 1

A

B

C

D

E

F

G

	ice_cream_sales_dataset						
1	Temperature	Month	isWeekend	isHoliday	PreviousDaySales	Price	Sales
2	35.6	Jan	0	1	72.0	3.27	54.5
3	34.4	Jan	0	0	54.5	3.4	45.1
4	29.6	Jan	0	0	45.1	3.32	20.0
5	26.8	Jan	1	0	20.0	3.13	31.0
6	26.2	Jan	1	0	31.0	3.3	28.2
7	25	Jan	0	0	28.2	3.27	5
8	28.9	Jan	0	0	5	3.28	11.7
9	37.8	Jan	0	0	11.7	3.22	58.7
10	32.0	Jan	0	0	58.7	3.18	32.5
11	34.4	Jan	0	0	32.5	3.28	9.9
12	25	Jan	1	0	9.9	2.93	38.1
13	37.3	Jan	1	0	38.1	2.9	58.8
14	25	Jan	0	0	58.8	3.2	6.6
15	26.7	Jan	0	0	6.6	3.28	10.6
16	27.9	Jan	0	0	10.6	3.31	24.8
17	37.3	Jan	0	0	24.8	3.19	34.7
18	34.0	Jan	0	0	34.7	3.33	30.3
19	29.6	Jan	1	0	30.3	3.32	58.4
20	31.7	Jan	1	0	58.4	3.33	62.3
21	36.9	Jan	0	1	62.3	3.25	87.4
22	25	Jan	0	0	87.4	3.32	9.3
23	38.5	Jan	0	0	9.3	3.23	41.3
24	41.2	Jan	0	0	41.3	3.21	45.3
25	29.8	Jan	0	0	45.3	3.32	31.1

Actual

54.5